Clustering and Customer Segmentation

EventEdge Corporate Training: Participant Segmentation Using K-means Clustering

Prepared for: Assignment 5 - Unsupervised Learning for Pattern Discovery and Customer/User Segmentation
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Model Used: K-means Clustering
Reference Outcome: High_Dropout_Risk (not used for clustering)
Number of Clusters: K = 4

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1. Executive Summary

This report presents Assignment 5 clustering and customer segmentation project for EventEdge Corporate Training. The project uses unsupervised learning to discover participant groups based on course characteristics, first-week engagement, reminder-email activity, and satisfaction. The report connects the technical clustering results to business decision-making, participant support planning, marketing strategy, responsible AI use, and practical management recommendations.

The original dataset contained 380 rows and 19 columns. No duplicate rows were found, so the final cleaned dataset also contained 380 rows and 19 columns. Two columns contained missing values before cleaning: Manager_Support_Level had 6 missing values and Satisfaction_Score_Week1 had 9 missing values. Missing values were handled using simple and transparent rules before clustering.

The final K-means model used K = 4. The four segments identified were Active Short-Course Learners, At-Risk Long-Course Low-Engagement Participants, Premium Engaged Long-Course Participants, and Moderate-Engagement Short-Course Participants. The most valuable group appears to be the Premium Engaged Long-Course Participants segment, while the segment needing the most marketing and support attention is the At-Risk Long-Course Low-Engagement Participants group.

Because this is an unsupervised learning assignment, High_Dropout_Risk was not used as a target variable for model training. It was reviewed only as a reference outcome to support business interpretation after the clusters were created.

2. Introduction and Assignment Objective

The objective of this assignment is to apply unsupervised learning to discover meaningful groups in a marketing-related dataset. Unlike a supervised classification assignment, the goal is not to predict a Yes or No outcome. Instead, K-means clustering is used to find hidden participant segments based on similar patterns in numerical features.

This report follows a structured workflow from business understanding and data preparation to feature selection, scaling, K-means modelling, cluster interpretation, visualization, business recommendations, and responsible AI reflection. The structure is similar to previous supervised learning reports, but the modelling logic is adjusted for unsupervised learning.

3. Business Context and Machine Learning Decision

EventEdge Corporate Training is a B2B learning company that sells professional development programs to client organizations. Its courses include Data Analytics, Leadership, Project Management, Customer Experience, Digital Marketing, and AI for Business. Client companies pay for employees to participate and expect stronger skills, higher productivity, and measurable learning outcomes.

The main business problem is participant dropout. When employees stop attending or fail to complete a course, EventEdge risks lower client satisfaction, reduced renewal rates, wasted coaching capacity, and weaker training outcomes. The business therefore needs a practical way to understand participant groups and match support strategies to different engagement patterns.

The machine learning decision supported by this clustering model is: Which participant groups should EventEdge prioritize for different marketing, communication, and support actions?

Table 1: Business challenge and possible segmentation action

Business Challenge	Why It Matters	Possible Segmentation Action
Low first-week engagement	Participants with weak early attendance or LMS activity may disengage before completing the course.	Create early check-in workflows, reminders, and coaching support.
Long-course participation risk	Longer courses may require more sustained motivation and milestone-based support.	Use milestone messages, progress tracking, and manager reminders.
Low satisfaction signals	Low week-1 satisfaction can indicate confusion, mismatch, or early frustration.	Provide service recovery, course guidance, or advisor follow-up.
Limited manager support	Corporate training participants may need workplace encouragement to complete assigned programs.	Send manager-support prompts and progress summaries to client contacts.
Different participant values	Some segments may represent higher-value or premium course participants.	Protect high-value relationships through personalized support and reporting.

4. Dataset Overview

The dataset used for this project is eventedge_training_dropout_risk_dataset.csv. Each row represents one participant enrolled in an EventEdge corporate training course. The dataset includes client context, participant context, course details, early engagement signals, support signals, and a reference dropout-risk outcome.

Table 2: Dataset and project summary

Item	Value
Original dataset shape	380 rows x 19 columns
Duplicate rows before cleaning	0
Duplicate rows after cleaning	0
Final cleaned dataset shape	380 rows x 19 columns
Reference outcome	High_Dropout_Risk
Reference outcome use	Reviewed after clustering only; not used to train K-means
Problem type	Unsupervised learning / clustering
Model used	K-means clustering
Final number of clusters	K = 4

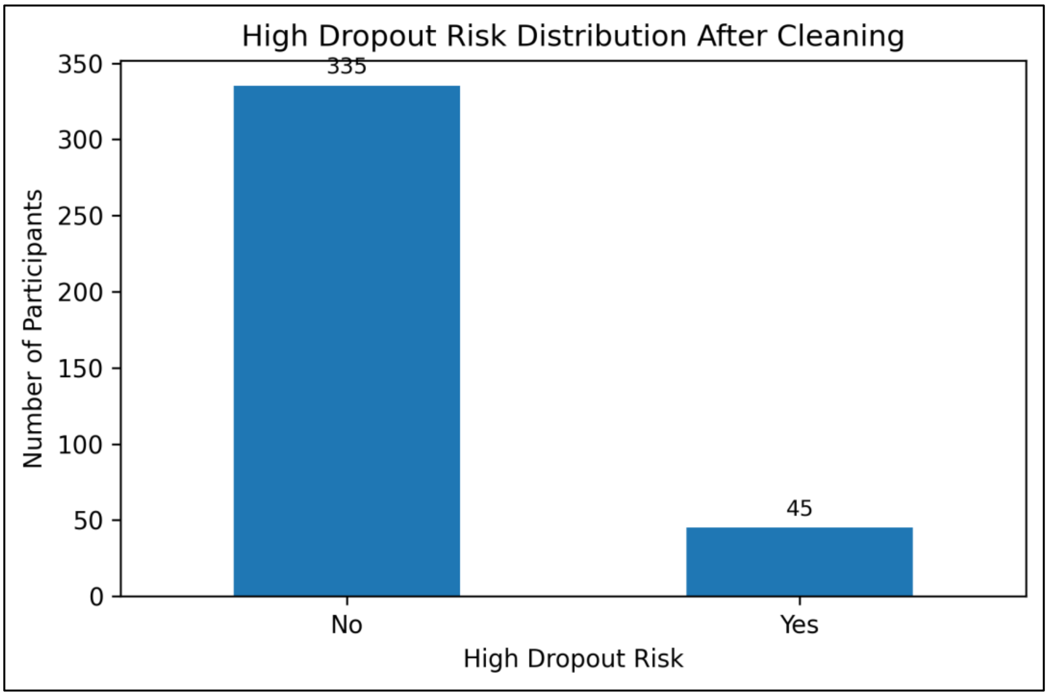

Figure 1: High dropout risk distribution after cleaning

Figure 1 shows that the cleaned dataset contains 335 participants marked No and 45 participants marked Yes for High_Dropout_Risk. This distribution is useful for business context, but the column is not used as a target variable in the clustering model.

5. Reference Outcome and Input Features

The dataset contains High_Dropout_Risk, where Yes indicates that a participant is marked as high dropout risk and No indicates that the participant is not marked as high dropout risk. However, because this assignment is based on unsupervised learning, High_Dropout_Risk is treated only as a reference outcome and is not used to train the model.

The clustering model uses selected numerical variables that describe course length, fee level, enrollment timing, first-week attendance, LMS activity, reminder-email engagement, and first-week satisfaction. Participant_ID is excluded because it is an identifier, and High_Dropout_Risk is excluded because it is a reference outcome.

Table 3: Dataset columns and business meaning

Column	Type	Business Meaning
Participant_ID	Identifier	Unique participant code; excluded from clustering.
Enrollment_Date	Categorical/ Date	Date when the participant enrolled in the course.
Industry	Categorical	Client organization industry.
Company_Size	Categorical	Size category of the client company.
Job_Level	Categorical	Participant job level or seniority.
Enrollment_Channel	Categorical	Channel through which the participant enrolled.
Course_Type	Categorical	Type of corporate training course.
Course_Length_Weeks	Numerical	Length of the training course in weeks.
Fee_Paid	Numerical	Fee paid for the participant or course enrolment.

Column	Type	Business Meaning
Days_Between_Enrollment_and_Start	Numerical	Number of days between enrolment and course start.
Sessions_Attended_First_Week	Numerical	Number of sessions attended during the first week.
LMS_Logins_First_Week	Numerical	Learning management system logins during the first week.
Assignment1_Submitted	Categorical	Whether the first assignment was submitted.
Manager_Support_Level	Categorical	Level of manager or workplace support.
Prior_Training_Completed	Categorical	Whether prior training had been completed.
Workload_Level	Categorical	Participant workload level.
Reminder_Emails_Opened	Numerical	Number of reminder emails opened.
Satisfaction_Score_Week1	Numerical	First-week satisfaction score.
High_Dropout_Risk	Reference outcome	Reference dropout-risk indicator; not used for clustering.

Table 4: Numerical and categorical feature groups

Feature Type	Number of Features	Columns
Numerical features	7	Course_Length_Weeks, Fee_Paid, Days_Between_Enrollment_and_Start, Sessions_Attended_First_Week, LMS_Logins_First_Week, Reminder_Emails_Opened, Satisfaction_Score_Week1
Categorical features	10	Enrollment_Date, Industry, Company_Size, Job_Level, Enrollment_Channel, Course_Type, Assignment1_Submitted, Manager_Support_Level, Prior_Training_Completed, Workload_Level

6. Data Preparation and Data Quality Review

Data preparation was completed before clustering to improve data quality and make the dataset suitable for K-means. The inspection included checking the dataset shape, column names, data types, missing values, duplicate rows, reference outcome distribution, numerical features, and categorical features.

Table 5: High dropout risk distribution before and after cleaning

Stage	No	Yes	Total
Before duplicate removal	335	45	380
After duplicate removal	335	45	380

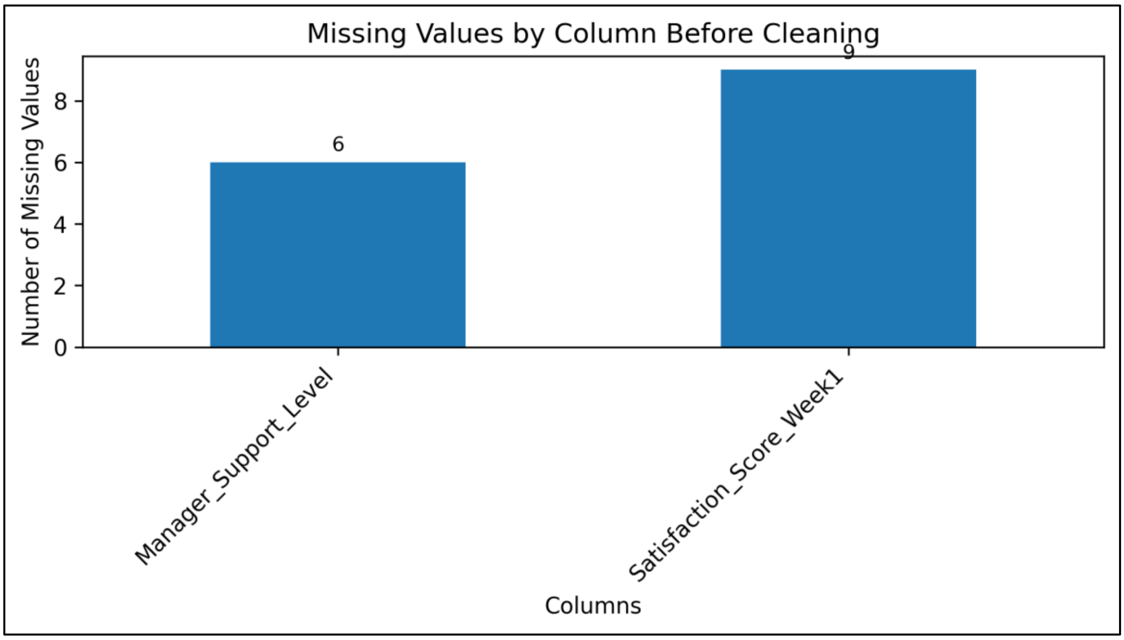

Figure 2: Missing values by column before cleaning

Figure 2 shows that missing values appeared in Manager_Support_Level and Satisfaction_Score_Week1. These were handled before clustering so that the selected numerical features could be scaled and used by K-means.

Table 6: Missing value inspection and handling plan

Column	Data Type	Missing Values	Planned Method	Handling Location
Manager_Support_Level	Categorical	6	Most frequent value	Before clustering
Satisfaction_Score_Week1	Numerical	9	Median	Before clustering

Because this is an unsupervised learning assignment, there is no train-test split. The objective is to segment the full dataset rather than evaluate prediction performance on a held-out test set. Therefore, missing-value handling was completed before clustering. This does not create the same data leakage issue as supervised learning, but a future production workflow should apply the same fitted preprocessing rules consistently to new participants.

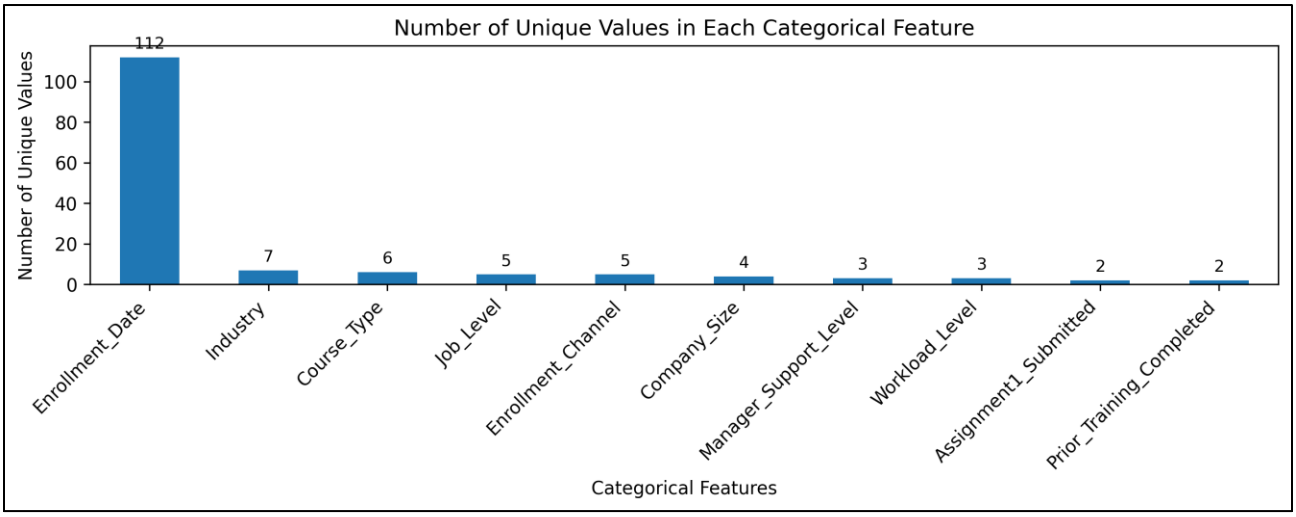

Figure 3: Number of unique values in each categorical feature

Figure 3 summarizes the number of unique values in each categorical feature. These categorical variables support business understanding but are not directly used in the final K-means model, which uses selected numerical features.

7. Feature Selection and Scaling for Clustering

K-means clustering is a distance-based algorithm, so feature selection and scaling are important. The selected features should describe participant behaviour and course context, while variables with very different ranges must be scaled so that one large-scale variable does not dominate the clustering result.

Table 7: Selected numerical features for clustering

Selected Feature	Business Meaning
Course_Length_Weeks	Course duration and complexity
Fee_Paid	Business value and course price level
Days_Between_Enrollment_and_Start	Enrollment timing before course start
Sessions_Attended_First_Week	Early attendance and participation
LMS_Logins_First_Week	First-week digital engagement
Reminder_Emails_Opened	Response to communication and reminders
Satisfaction_Score_Week1	Early satisfaction and learning experience

The selected features were scaled using StandardScaler before applying K-means. Scaling transforms the variables to a comparable scale, which helps ensure that the clustering result is not driven mainly by high-value variables such as Fee_Paid.

8. Elbow Method and K-means Modelling

The Elbow Method was used to help choose a suitable number of clusters. The method compares K-means inertia values for different values of K. Inertia generally decreases as K increases, but the goal is to choose a K value that balances technical separation with practical business interpretation.

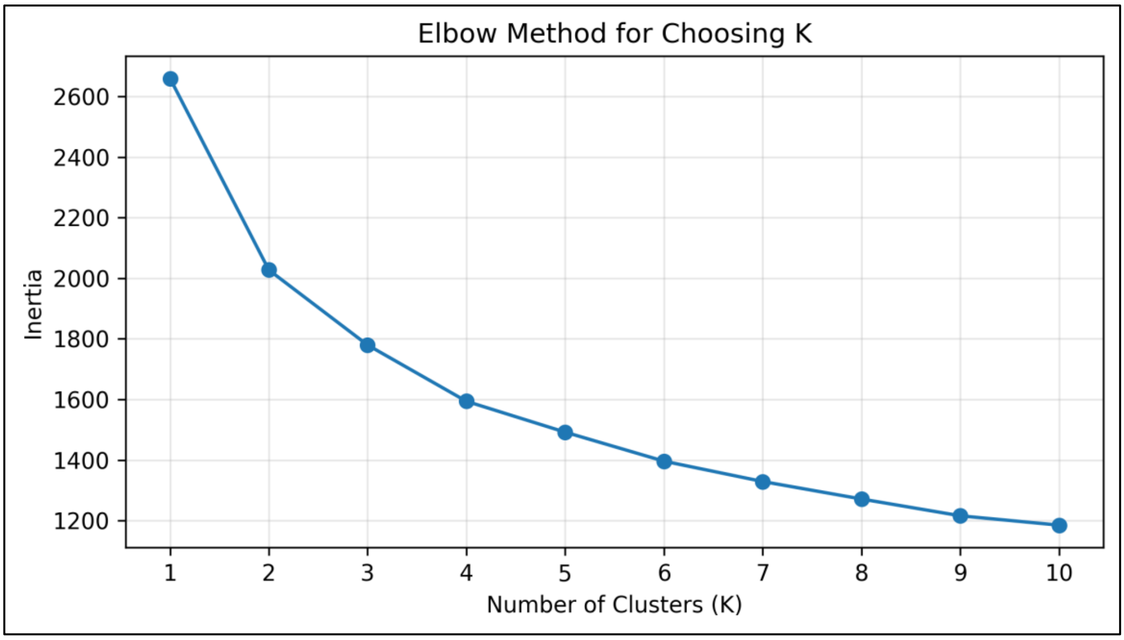

Figure 4: Elbow Method for choosing K

The elbow curve decreases quickly at first and then begins to flatten. The elbow curve does not show a perfectly sharp turning point, but K = 4 provides a reasonable balance between lower inertia and business interpretability. Four clusters allow EventEdge to describe participants in practical business language and create meaningful participant segments for marketing and support decisions.

Table 8: Elbow Method inertia values

K	Inertia
1	2660.00
2	2027.46
3	1779.56
4	1593.86
5	1491.90
6	1396.11
7	1328.90
8	1271.16
9	1215.76
10	1184.86

Table 9: K-means model configuration

Model Item	Value
Algorithm	KMeans
Number of clusters	4
Random state	42
n_init	10
Input data used	Scaled selected numerical features
Input shape after scaling	380 rows x 7 features
Target variable	None; unsupervised learning

9. Cluster Analysis and Segment Profiles

After training the final K-means model, cluster labels were added back to the cleaned dataset. Each cluster was given a practical business name based on its average feature values and business interpretation.

Table 10: Segment counts

Cluster	Segment Name	Number of Participants
0	Active Short-Course Learners	82
1	At-Risk Long-Course Low-Engagement Participants	94
2	Premium Engaged Long-Course Participants	115
3	Moderate-Engagement Short-Course Participants	89

Table 11: Cluster profile - course and business context

Cluster	Segment Name	Course Length Weeks	Fee Paid	Days Before Start
0	Active Short-Course Learners	5.88	$1425.12	13.43
1	At-Risk Long-Course Low-Engagement Participants	9.68	$2055.03	10.90
2	Premium Engaged Long-Course Participants	10.19	$2085.31	12.81
3	Moderate-Engagement Short-Course Participants	5.75	$1306.19	14.96

Table 12: Cluster profile - engagement and satisfaction

Cluster	Segment Name	Sessions First Week	LMS Logins First Week	Reminder Emails Opened	Satisfaction Week 1
0	Active Short-Course Learners	2.90	7.76	3.68	4.37
1	At-Risk Long-Course Low-Engagement Participants	1.02	2.49	1.13	2.59
2	Premium Engaged Long-Course Participants	2.28	6.38	3.13	4.20
3	Moderate-Engagement Short-Course Participants	1.26	4.97	1.64	3.25

Table 13: High dropout risk distribution by segment

Segment	No Count	Yes Count	Total	Yes Percentage
Active Short-Course Learners	82	0	82	0.00%
At-Risk Long-Course Low-Engagement Participants	63	31	94	32.98%
Premium Engaged Long-Course Participants	112	3	115	2.61%
Moderate-Engagement Short-Course Participants	78	11	89	12.36%

Cluster 0, Active Short-Course Learners, has shorter courses, lower fees, strong first-week attendance, high LMS logins, strong reminder-email engagement, and high satisfaction. This group appears stable and engaged.

Cluster 1, At-Risk Long-Course Low-Engagement Participants, has longer courses, higher fees, low attendance, low LMS activity, fewer reminder emails opened, and the lowest satisfaction. This group has the highest observed percentage of High_Dropout_Risk = Yes and needs the most support attention.

Cluster 2, Premium Engaged Long-Course Participants, has longer courses, higher fees, strong LMS activity, good satisfaction, and low observed dropout-risk levels. This group appears to be the most valuable business segment because it combines engagement and higher course value.

Cluster 3, Moderate-Engagement Short-Course Participants, has shorter courses and moderate engagement. This group is not as concerning as Cluster 1, but it may benefit from light reminders and engagement nudges.

10. Cluster Visualizations

Visualizations help connect the clustering results to practical business interpretation. The following charts show segment sizes, engagement differences, two-feature separation, and PCA-based two-dimensional cluster visualization.

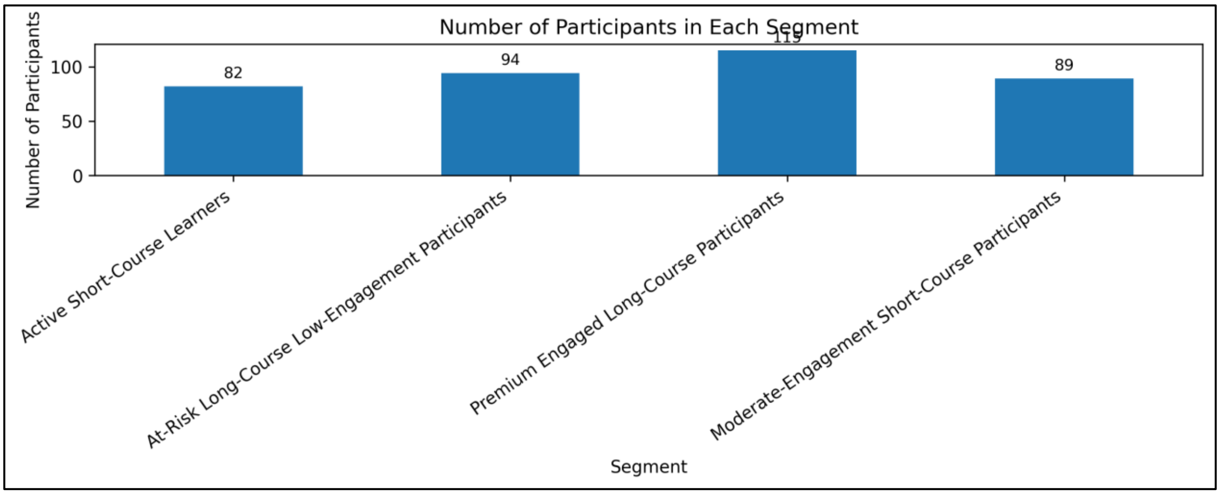

Figure 5: Number of participants in each segment

Figure 5 shows that all four segments are large enough to support practical analysis. The largest segment is Premium Engaged Long-Course Participants, while Active Short-Course Learners is the smallest segment.

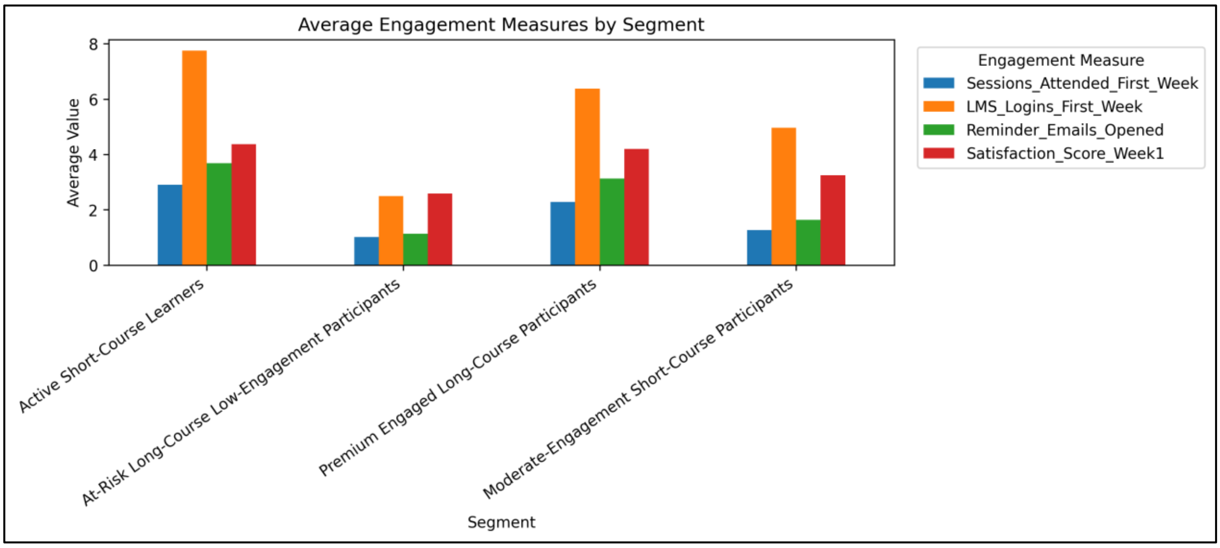

Figure 6: Average engagement measures by segment

Figure 6 shows clear differences in first-week attendance, LMS logins, reminder-email openings, and satisfaction across segments. This supports the business interpretation that engagement and satisfaction are central to participant segmentation.

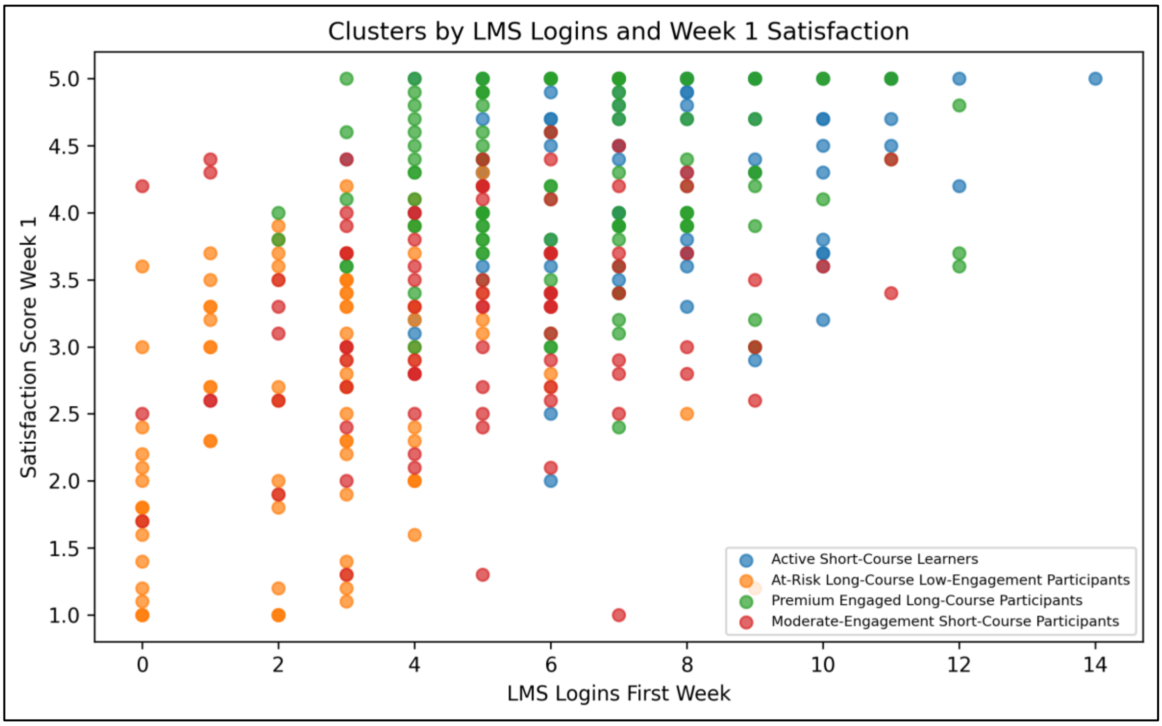

Figure 7: Clusters by LMS logins and week 1 satisfaction

Figure 7 compares participants using LMS logins and first-week satisfaction. These two features are useful because they show early engagement and participant experience. The at-risk segment is concentrated around lower LMS activity and lower satisfaction.

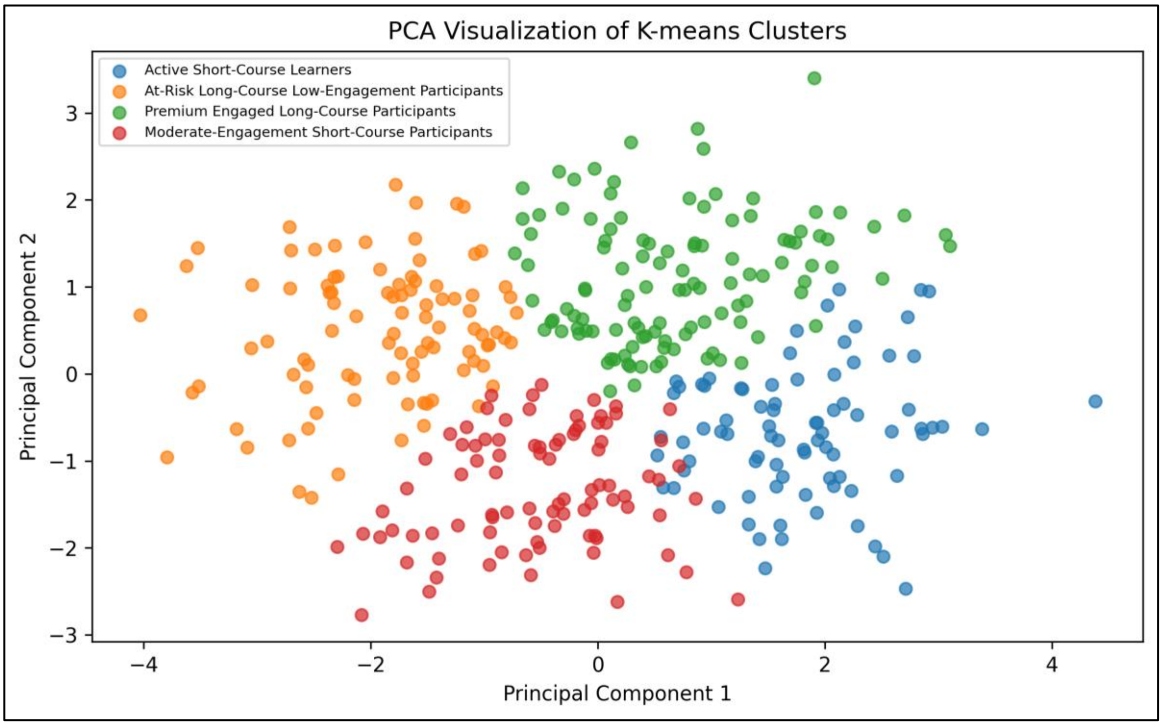

Figure 8: PCA visualization of K-means clusters

Figure 8 reduces the scaled clustering features into two principal components so that the clusters can be displayed in two dimensions. PCA does not replace the full K-means model, but it gives a useful visual overview of the segment separation. Some overlap between clusters is expected because PCA compresses seven scaled features into only two dimensions.

11. Business Recommendations

The clustering results can help EventEdge move from a one-size-fits-all participant support approach to a more targeted segmentation strategy. Each segment should receive support that matches its behaviour and business value.

Table 14: Business recommendations by segment

Segment	Business Interpretation	Recommended Action
Active Short-Course Learners	High engagement and high satisfaction in shorter courses.	Maintain standard support, encourage testimonials, and use light engagement messages.
At-Risk Long-Course Low-Engagement Participants	Low attendance, low LMS activity, low email response, and lowest satisfaction.	Prioritize early check-ins, coaching, manager support, flexible deadlines, and onboarding improvement.
Premium Engaged Long-Course Participants	Higher course value, longer courses, strong engagement, and strong satisfaction.	Protect this segment with premium service, progress reporting, and relationship management.
Moderate-Engagement Short-Course Participants	Moderate activity and satisfaction in shorter courses.	Use reminder nudges, progress messages, and simple support prompts to improve completion.

The most important pattern discovered is that participant groups differ strongly by early engagement and satisfaction. Participants with strong LMS activity, good attendance, and high satisfaction appear more stable, while low-engagement participants in longer courses may need more immediate support. EventEdge can use these segments to improve marketing messages, client reporting, onboarding design, and support prioritization.

12. Limitations, Bias, and Responsible AI

K-means clustering is useful for discovering patterns, but it does not produce perfect participant segments. The algorithm groups records based on distance, so results can be affected by feature selection, scaling, outliers, and the selected number of clusters.

Table 15: Limitations and mitigation plan

Limitation	Business Risk	Mitigation / Future Improvement
Synthetic and simplified dataset	The data may not fully represent real participant behaviour or all causes of dropout.	Validate the analysis using real historical EventEdge data before operational use.
K-means assumptions	Clusters may not be perfectly separated or may not match all real business groups.	Use cluster results as a decision-support tool, not as final truth.
Choice of K	The elbow curve is not perfectly sharp, so K = 4 is a practical choice rather than a guaranteed perfect answer.	Review alternative K values and confirm that segment names remain useful to managers.
Possible bias by job level, industry, or company size	Some groups could receive different levels of attention if clusters are used without review.	Monitor segment outcomes across industries, job levels, and company sizes.
Reference outcome not used in model	High_Dropout_Risk supports interpretation but does not define the clusters.	Use it only as business context and continue to apply human judgment.

Responsible use requires human review. A participant in a lower-engagement segment should not be blamed or treated unfairly. Instead, the cluster should trigger helpful and respectful actions such as reminders, coaching, flexible deadlines, or manager communication. Program advisors should consider participant context before making decisions.

13. Conclusion

This assignment applied K-means clustering to discover participant segments for EventEdge Corporate Training. The workflow included business understanding, data inspection, duplicate and missing-value review, data cleaning, numerical feature selection, scaling, Elbow Method analysis, final K-means modelling, cluster profiling, visualizations, business recommendations, and responsible AI reflection.

The final model identified four participant segments: Active Short-Course Learners, At-Risk Long-Course Low-Engagement Participants, Premium Engaged Long-Course Participants, and Moderate-

Engagement Short-Course Participants. These segments help EventEdge understand that different participant groups require different marketing and support strategies.

Overall, the clustering analysis provides a useful starting point for participant segmentation. It can support proactive intervention, better communication, improved onboarding, and stronger client reporting. However, the results should support human judgment rather than replace it.

Appendix A: Assignment Alignment Checklist

Table 16: Assignment alignment checklist

Assignment Requirement	Where Addressed in This Report
Understand the business problem	Sections 3 and 11 explain participant dropout, business risk, and segmentation-based support actions.
Prepare the dataset	Sections 4 to 7 document data inspection, duplicate rows, missing values, cleaning, feature groups, and scaling.
Apply K-means clustering	Section 8 explains the Elbow Method and K-means model configuration.
Analyze and interpret clusters	Section 9 provides segment counts, cluster profiles, dropout-risk reference distribution, and cluster interpretation.
Visualize the clusters	Section 10 includes the required visualizations and interpretation.
Business interpretation	Section 11 provides segment-level recommendations and decision-making value.
Limitations and responsible AI reflection	Section 12 discusses limitations, bias, fairness, and human judgment.